

Errors

Hypothesis Testing

A systematic framework for evaluating whether the evidence in our data is strong enough to reject a specified null hypothesis.

Types of Errors:

Type I Error

False Positive

Incorrectly reject a null hypothesis that is actually true

Type II Error

False Negative

Incorrectly fail to reject a null hypothesis that is actually false

Choosing a Significance Level

Significance level (α): The probability of a Type I error when the null hypothesis is true.

A significance level of 0.05 means that if H_0 is true, we will incorrectly reject it about 5% of the time.

Reject H_0 when the p-value is below the significance level.

Setting Up Tests

Statistical power: Probability of correctly rejecting H_0 when H_0 is false.

- Statistical power increases with sample size.
- $\text{Power} = 1 - \text{probability of a Type II error}$

Knowledge Check

Suppose male and female Gentoo penguins actually do not have different mean flipper lengths, but our statistical test concludes that they do. **What type of error did we make?**

- a) Type I Error
- b) Type II Error

Knowledge Check

Suppose male and female Gentoo penguins actually do have different mean flipper lengths, but our statistical test fails to detect a difference. **What type of error did we make?**

a) Type I Error

b) Type II Error



University of California
San Francisco